

Reading the Current

What three months of Geneformer work taught us
about believing a model's answer



Kaisar Dauyey · Shinji Nakaoka

Laboratory of Mathematical Biology, Faculty of Advanced Life Science, Hokkaido University

TODAY

The model always answers. We spent three months learning when the answer means something.

I

The chart

Geneformer and in-silico perturbation, from zero.

II

How we got here

July to September, including what we got wrong.

III

Rules we learned

Six checks we now apply before believing a result.

IV

This week

The same design, tried in colorectal cancer.

V

Join us

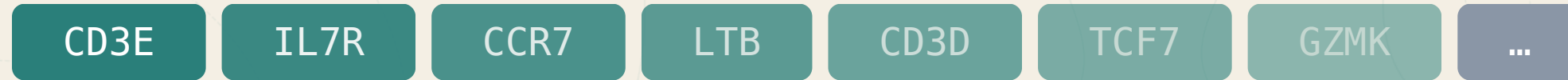
Open questions and small tasks.

Unfamiliar words are collected in a glossary on slide 27 and on the handout.

I The chart

Geneformer for beginners

Geneformer reads each cell as a ranked list of genes



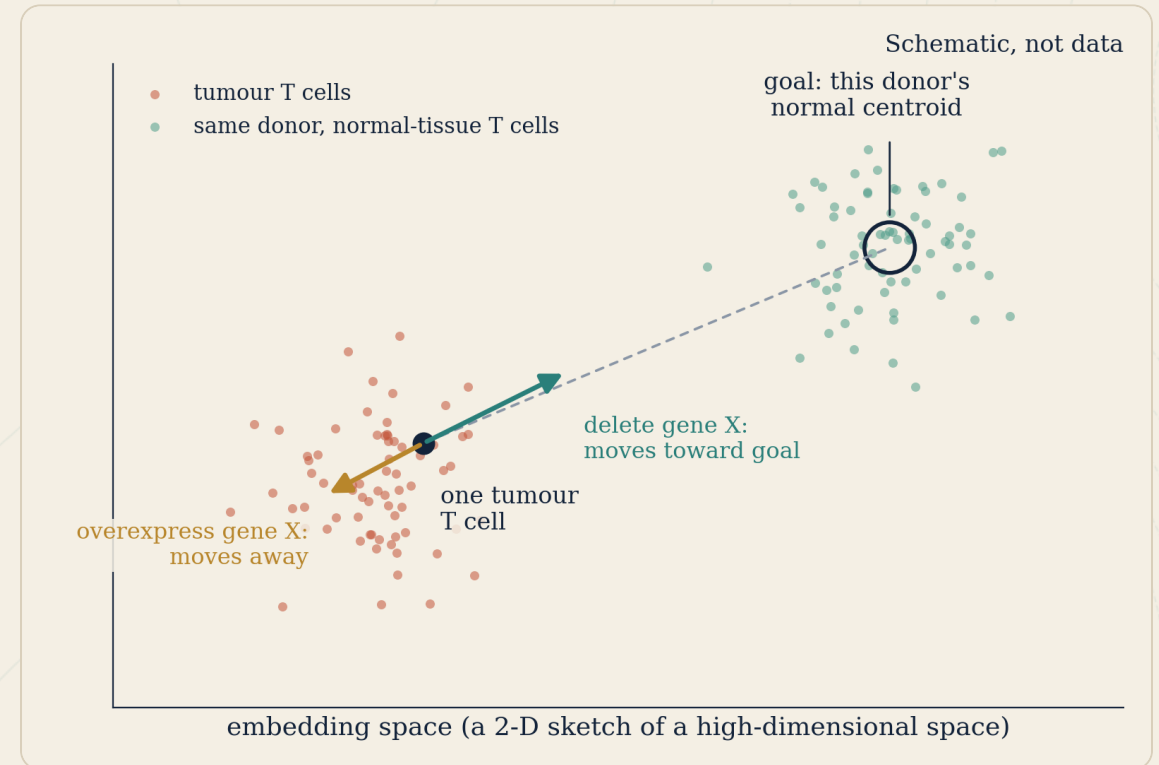
An illustrative T cell, written as genes ordered from most to least characteristic. Each gene in the list is a **token**, the model's word. Not real data.

The model divides each gene's expression by how high that gene usually is, then sorts. Housekeeping genes sink and the genes that make this cell distinctive rise to the front. A transformer (a neural network that reads sequences) then learns to guess hidden genes from the rest of the list, across tens of millions of cells; the V2 models saw about 104 million cells. What we keep is the model's internal position for each cell, its **embedding**: a point on the chart.

A compass for our question, then a nudge

We first **fine-tune** the model on one distinction: tumour T cells against the same patient's normal-tissue T cells. Then we edit a cell's sentence. **Deleting** a gene removes its token; **overexpressing** it moves the token to the front. We measure whether the cell moved toward the centre of that patient's normal T cells: the change in similarity to that goal (the cosine shift; positive means closer to normal).

In-silico perturbation (ISP) is a question we put to the model, not an experiment on cells.



II

How we got here

July to September, including what we got wrong

It began with a screen, and an abstract we had to support

The July screen

21,000 lung T cells in three classes: adenocarcinoma (LUAD), squamous carcinoma (LUSC) and normal tissue, using the smaller Geneformer model (104 million parameters). Accuracy 0.7834, and 2,937,776 held-out deletions of single genes in single cells.

Our own audit, 28 July

"The present LUAD/LUSC exploratory work cannot by itself substantiate the abstract's SCLC claims."

So we built a small-cell lung cancer (SCLC) line: 46,140 T cells from 42 patients (we use "donor" and "patient" for the same thing), held-out accuracy 0.919. The normal test class was a single patient, 566 cells.

What we told the JSDP meeting in August

- We deleted and overexpressed 50 genes. Four genes passed in both directions in all three SCLC test patients: deleting them moved cells toward normal and overexpressing moved them away. They were TIM-3 (HAVCR2), TIGIT, CTLA-4 and IL7R, all linked to T-cell exhaustion or persistence. (Slide 18 shows why that pattern alone is not enough.)
- In independent SCLC tissue (spatial transcriptomics), an antigen-presentation programme the model pointed to tracked T-cell abundance: $p = 0.361$, 7.95 standard deviations above a null of 100 random gene sets matched on mean expression. The broader dysfunction benchmark gave $p = 0.161$.
- The poster flagged one thing as open. Along an axis built from immune-checkpoint genes, the model placed the states as Normal < SCLC < LUAD, which is not what we expected.

We closed the talk with a question: "if SCLC T cells are not exhausted, what are they, and why does ICI still fail?" (ICI: immune checkpoint inhibitors.)

Then we tested our own story, and it did not hold

Not a line

The three cell states form a triangle in the model's space (SCLC interior angle 61.94°). So there is no single exhaustion axis to move along.

Inside the null

Overexpressing an exhaustion gene programme in SCLC cells moved them toward LUAD by $+0.0873$. Expression-matched random gene sets did almost as much, $+0.0816$ on average; one-sided $p = 0.4286$. Our shift was no bigger than random genes give.

One patient

An apparent conflict between two analyses came from one SCLC patient holding 74.9% of the SCLC test cells. Counting patients instead (19 vs 22): $p = 0.216$.

Our strongest hits were not what they looked like

The top of the screen

S100A8 had a false discovery rate of 1.45×10^{-224} . Yet across S100A8 and S100A9 in six state-to-state comparisons (12 rows), deleting and overexpressing moved cells the *same* way in 9. A dose-responsive driver cannot do that.

Of the top 120 rows, 55 were flagged as likely ambient RNA from surrounding tissue; none was one of our T-cell anchor genes.

The July "normal" class

None of the 18 studies contributed both normal and tumour T cells. "Normal versus tumour" had partly been "these studies versus those studies".

The July adenocarcinoma class was also 27.4% metastatic tissue.

We found two faults in our own instrument

Two arms, two cell sets

The August panel overexpressed each gene into all 2,424 SCLC test cells but deleted it only in cells that carried it (202 to 1,131). The arms were scored on different cells.

We added a caveat to the four August hits, "a qualifier, not a retraction". The panel has not been rerun.

A silent default

When given a list of genes, Geneformer's own code inserts an overexpressed gene into every cell, whether or not the cell expressed it: 14,738 of 15,179 calls in our rebuilt study.

We had written this down on 22 September and missed it three days later. Our amendment says so: "Prior art in this repository, missed."

So we rebuilt the experiment, and wrote the rules down

- **A cleaner design:** 43 adenocarcinoma patients, each with tumour and normal T cells, 100 cells per tissue, the larger Geneformer model (316 million parameters). The classifier reached 0.825 balanced accuracy, with all 43 patients above chance.
- **The July list fell away:** 13 of its 15 genes could not be tested in T cells at all (keratins, a mucin, a haemoglobin chain). Of 34 testable T-cell genes, 13 gave stable calls.
- **A warning sign:** even the matched control genes showed deletion and overexpression shifts anti-correlated, $\rho = -0.62$.
- **28 September:** the lab adopted a written standard for judging ISP (ISP-STD-1). Its worked example asked whether S100 genes with high ambient-RNA risk beat their matched controls more than low-risk S100 genes do. The answer was negative in both contrasts ($p = 6/70$ and $34/70$).

III

Rules we learned

How we now judge an ISP result (ISP-STD-1)

File the voyage plan before you sail

Before any output exists, we write down the test, its predicted direction, whether it is one- or two-sided, the significance threshold (alpha), and the **smallest p-value the test can actually reach** at our sample size. We also fix the controls, the budget and the words we will use for each outcome. Someone independent checks it, and the sign-off names file hashes.

Why: in August we read meaning into an ordering before we had a test for it.

FROM THIS WEEK

The colorectal (E2) test uses 13 lung reference genes, of which 10 are **expected** to be testable in colon, so the expected $n = 10$. At $n = 10$ a one-sided binomial test needs **at least 9 of 10** genes to agree ($P = 11/1024 = 0.011$; 8 of 10 gives 0.055). If n ends at 9, the bar is 8 of 9. We wrote the rule down before any number existed.

The first review of the E2 plan was not passed: it claimed an unchanged software environment, but pandas had been downgraded since the lung runs. The record was corrected before the GPU started.

Check the compass, then check that an anchored boat stays still

Classifier gate

If the fine-tuned model cannot tell tumour from normal in patients it never saw, there is no axis to measure along.

Pass: pooled held-out balanced accuracy ≥ 0.60 and an exact two-sided sign test over patients, $p \leq 0.05$. Fail: stop before any perturbation.

Why: the July classifier could not be told apart from a study classifier.

No-op gate

Run the pipeline twice on identical input with no edit, as two independent forward passes.

Pass: every per-cell shift is exactly 0, or within measured noise.

Why: in gene-list mode, our code once overexpressed into every cell without our asking (14,738 of 15,179 calls).

Compare against boats of the same build

Each tested gene is compared with at least 20 control genes matched within $0.5 \log_2$ on detection and within 5 rank-percentile points. If we cannot find 20, the gene is *not estimable*; we never widen the tolerances. Patients, not cells, are the unit. A result must survive dropping any single control, and hold in at least 95% of 10,000 bootstrap draws.

Why: S100A8 topped the screen on significance alone, and one patient once carried three quarters of the SCLC test cells.

WHY MATCHING MATTERS

Highly detected genes move cells more, whatever they do biologically. A gene is only interesting if it moves cells *more than genes that are detected as often*.

In the lung panel, 318 control genes were matched to the curated genes.

III · RULE 5 · THE CURRENT

Every boat drifts. Measure the drift before claiming a nudge.

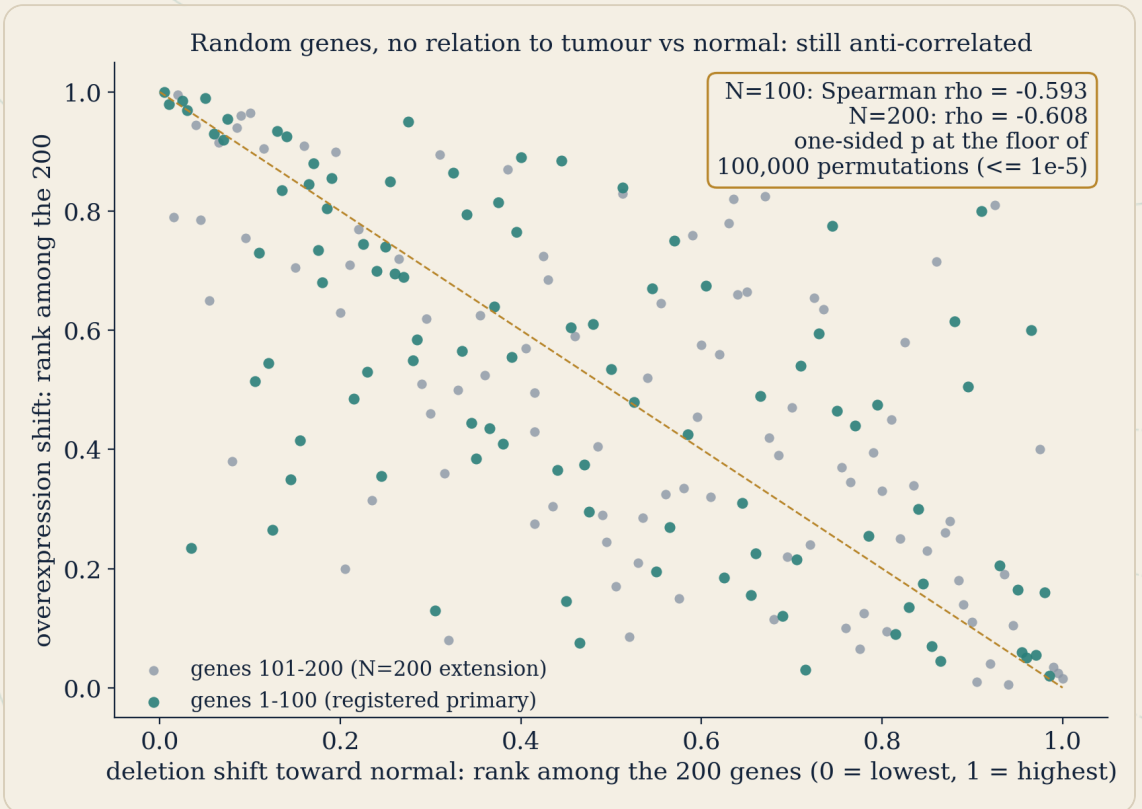
The control genes' -0.62 made us ask: what do **random genes** do? We ran the full pipeline on random genes with no known link to tumour versus normal (lung, 43 patients). Each dot is one random gene: x = how far deleting it moves cells toward normal, y = how far overexpressing it does (as ranks).

$$\rho = -0.593 \quad (N = 100)$$

$$\rho = -0.608 \quad (N = 200, \text{confirming})$$

One-sided p at the floor of 100,000 permutations ($\leq 1e-5$); stable in every leave-one-gene-out and all 10,000 bootstrap draws.

Why: even our matched control genes anti-correlated ($\rho = -0.62$).



Opposite signs for delete and overexpress are not evidence of specificity

What it shows

An anti-correlation between deletion and overexpression exists for genes in general, in this model and this design. It is part of the baseline.

A validity check on the study's own control genes (308 of the 318 had enough data) gave $\rho = -0.658$, the same sign as the -0.62 seen earlier. We compare the two by sign only.

What it does not show

It does not erase the panel result: 13 of 34 testable T-cell genes still beat their matched controls.

It narrows what that result can mean. Opposite signs alone are expected; an unusual *size* of effect, measured against this null, is the more specific evidence.

Now in the standard as clause B7. The random genes are "eligible genes detectable in LUAD tumour T cells", not the whole genome.

III · RULE 6

Write the log in a fixed vocabulary. Never "no effect".



Why: in August we called an unexplained ordering a finding, and later had to add "a qualifier, not a retraction" to our hits.

A **negative** result means the registered criterion was not met. We report it with its direction, p and n, because a small study can miss a real effect. A result in the wrong direction is labelled **opposite_direction** and never counted as support. A result that is right in direction but unstable across control draws stays **open**.

What ISP cannot show, however clean the result

Cause

A shift in embedding space is the model's geometry, not a mechanism. No gene is shown to regulate T-cell state.

Origin of the signal

The classifier may read T-cell state, ambient RNA from the surrounding tissue, or the CD4:CD8 mix. The gates cannot tell these apart.

Ground truth

Accuracy needs a real perturbation screen in the same cell type (standard section D). We do not yet have one for these T cells.

IV

This week

Does any of this hold outside lung?

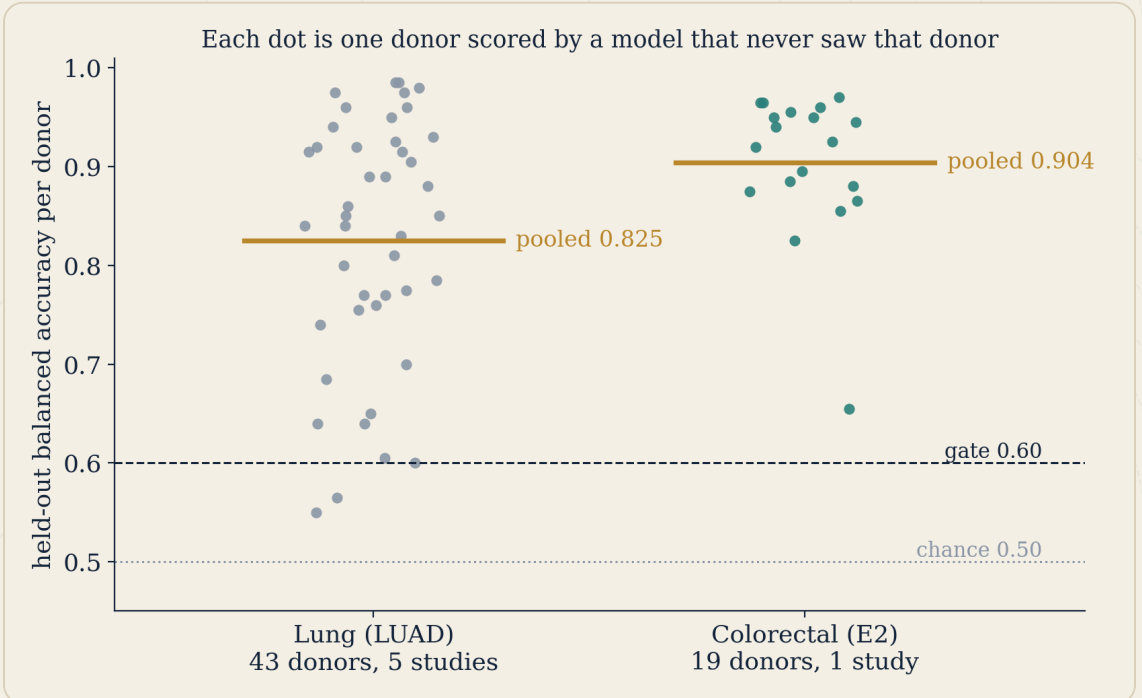
IV · DOES THE COMPASS WORK IN NEW WATERS?

Colorectal cancer (Pelka et al. 2021): the classifier gate passed

We took 19 patients whose tumour and normal samples were both unsorted, 100 T cells per patient per tissue, and fine-tuned **new models** from the base Geneformer-V2-316M, one per fold. These are not the lung models, and this is not zero-shot.

Pooled balanced accuracy was **0.904**, with **19 of 19** patients above chance (lowest 0.655) and a two-sided sign test $p = 1/262,144$. Folds ranged from 0.866 to 0.935, and the no-op gate's largest shift was 0.0.

This shows the tumour/normal split is learnable in colon. It does not show that the lung classifier transfers.



Colorectal perturbation screen: same direction as lung but not stable; the lung gene pattern did not replicate

What the run found (finished 10:20 JST, 2 October)

Random genes (primary). $\rho = -0.245$ over 100 genes, one-sided $p = 0.0067$; every leave-one-out test passes, but only 78% of 10,000 resamples stay significant, against the registered 95%. Status `control_draw_sensitive_open`: direction as in lung, not stable; open. Lung gave -0.593 .

Lung reference genes. 3 of 10 keep their lung direction, where 9 were needed ($p = 0.95$): `pattern_not_replicated`. One panel gene, PRF1, is dose-concordant in colon; it was open in lung.

369 genes $\times$ 19 patients $\times$ delete and overexpress; 32.6 GPU-hours of the 52 registered. A colon-lung difference cannot be put down to tissue alone.

What each registered outcome would tell us

Random genes (primary). Status positive: the anti-correlation appears in colon too, so it is more likely a property of the model and this design than of lung. negative (the anti-correlation did not appear) or `opposite_direction`: cannot be put down to tissue alone, since study, protocol, chemistry and the fine-tune all differ. `control_draw_sensitive_open`: stays open. **This is the outcome.**

Lung reference genes. `pattern_holds` if enough keep their lung direction (at $n = 10$, at least 9); `pattern_not_replicated` otherwise (**the outcome**); `not_testable` if fewer than 5 can be tested. Holding would not make any gene a regulator.

V

Uncharted waters

Open questions, and how to join

What we do not know yet

1. **Why does the current exist?** Is the delete/overexpress anti-correlation a property of rank encoding, of the model, or of lung biology? (The colorectal run gave a first answer: same direction as lung, but not stable.)
2. **Does the lung compass transfer?** Applying the frozen lung models to an independent lung cohort is planned, but the one candidate's data are not yet accessible to us.
3. **What does the classifier read?** T-cell state, ambient RNA, or CD4:CD8 composition?
4. **Where is ground truth?** No T-cell perturbation screen has yet been matched to our readout.
5. **Positive controls.** We have no registered genes known to drive this tumour/normal T-cell axis, so a negative result is hard to interpret.

Six ways in, from an hour to a month

1 AFTERNOON Re-derive a gate. Recompute the colorectal pooled balanced accuracy and sign-test p from `classifier_gate.json`. No GPU.

1 WEEK Propose positive controls. From the literature, list genes known to shift T cells between tumour and normal states, with sources.

2-4 WEEKS Explain the current. Test on synthetic sentences whether rank encoding alone produces the anti-correlation.

1 AFTERNOON Find the floor. For a planned n, compute the smallest attainable p of a sign test and of a Wilcoxon test, with ties. This is the attainability check.

1 WEEK Find ground truth. Survey published CRISPR or Perturb-seq screens in primary human T cells that could be matched to our readout.

1 HOUR Open a door. Help us check whether the independent lung cohort's data package (a Code Ocean capsule) contains per-cell metadata.

GLOSSARY

Words on the chart

Geneformer

A transformer pretrained on single-cell transcriptomes, each cell written as a ranked gene list.

Token

One gene in a cell's ranked list: the model's word.

Transformer

A neural network that reads sequences, here gene lists.

Donor

The patient a sample came from; we use donor and patient interchangeably.

Embedding

The model's internal coordinates for a cell; nearby cells look alike to the model.

Fine-tuning

Further training on one labelled task, here tumour vs normal T cells.

ISP

In-silico perturbation: delete or overexpress a gene token and measure the embedding shift toward a goal.

Cosine shift

The change in a cell's similarity to the goal after an edit; positive means closer to normal.

Goal centroid

The average embedding of a donor's own normal-tissue T cells.

Balanced accuracy

The mean of the true-positive and true-negative rates; 0.5 is chance.

Ambient RNA

Transcripts from surrounding tissue captured in a cell's droplet.

Matched controls

Genes chosen to resemble the tested gene in detection and rank, used as its comparison set.

Null control

The same pipeline on random genes, to measure the effect expected by default.

Pre-registration

Writing down the test, its threshold and its wording before any result exists.

Permutation floor

The smallest p a permutation test can report: $1/(\text{permutations} + 1)$.

The current is not the enemy. It is the baseline.

We learned to measure it first, then claim the nudge.

Sources for every number: SOURCES.md in this talk's folder · ISP-STD-1 · E2 registration (commit 8b11d5d)